

the image Penguins_grey.jpg:

```
A = imread('Penguins_grey.jpg');
B = fspecial('average');
C = filter2(B,A);
figure, imshow(A),figure, imshow(C/255)
```

Matrix B is a 3x3 matrix with each matrix element having a value of 0.111. Matrix C has data of type 'double.' We can divide it by 255 to obtain a matrix with values in the 0-1 range for use with imshow function. Both the original image and the filtered image are shown in Figs 1 and 2, respectively.

As you can see from the filtered image, the averaging filter blurs the image and the edges in the image are less distinct than in the original image. Larger the size of the averaging filter, more is the blurring effect. The code given below shows the averaging filter operation using an average filter of 9x9:

```
A = imread('Penguins_grey.jpg');
B = fspecial('average',[9,9]);
C = filter2(B,A);
figure, imshow(A),figure, imshow(C/255)
```

Fig. 3 shows the filtered image.

Averaging filter is an arithmetic mean filter. Other types of mean filters include geometric mean filters, harmonic mean filters and contra-harmonic mean filters. Mean filter is the simplest and the most widely used spatial smoothing filter.

The averaging filter operates on an $m \times n$ sliding window by calculating the average of all pixel values within the window and replacing the centre pixel value in the destination image with the result.

As mentioned above, the averaging filter causes the blurring effect in the image, proportional to the window size. This helps in reduction of noise, especially Gaussian or uniform noise. As an example, let us add salt and pepper noise to the Penguins_grey.jpg image and filter it using the averaging filter. Salt and pepper noise causes very bright (salt) and very dark (pepper) isolated spots in an image.

The command for adding salt



Fig. 5: The resultant image with salt and pepper noise removed

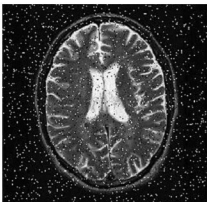


Fig. 6: Another image (MRI image) with salt and pepper noise

and pepper noise to an image is:

```
A = imread('Penguins_grey.jpg');
Noise_snp=imnoise(A,'salt & pepper');
imshow(Noise_snp)
```

Fig. 4 shows the Penguins_grey.jpg image with salt and pepper noise added.

Following commands remove the salt and pepper noise from Fig. 4:

```
>> B = fspecial('average',[9,9]);
>> C = filter2(B,Noise_snp);
>>imshow(C/255)
```

Fig. 5 shows the resultant image. As you can see, the image in Fig. 5 is blurred but the salt and pepper noise has been removed. Fig. 6 shows another image (MRI image) with salt and pepper noise and Fig. 7 shows the image after performing the following operations:

```
>> a=imread('MRI_snp.jpg');
>>imshow(a)
>> B = fspecial('average',[9,9]);
>> C = filter2(B,a);
>>imshow(C/255)
```



Fig. 7: The image in Fig. 6 after noise removal



Fig. 8: Gaussian filtered image with default standard deviation value of 0.5

A variation of the arithmetic mean filter is the geometric mean filter. It is primarily used on images with Gaussian noise. This filter is known to retain image detail better than the arithmetic mean filter.

Gaussian filters are another type of linear filter. These are low-pass filters, based on the Gaussian probability distribution function as given below:

$$f(x) = e^{-x^2/2\sigma^2}$$

where ' σ ' is the standard deviation.

Gaussian filters have a blurring effect which looks very similar to that produced by neighbourhood averaging.

The following commands apply Gaussian filter to the image Penguins_grey.jpg:

```
>> A = imread('Penguins_grey.jpg');
>> G1=fspecial('gaussian',[11,11]);
>> G1image=filter2(G1,A)/256;
>>imshow(G1image)
```

The numbers within the square